ZAID AHMAD KHAN

Chicago, IL | 773-980-8003 | zkhan41@hawk.iit.edu | Website | LinkedIn | GitHub
F-1 student, STEM OPT eligible (up to 3 years, no sponsorship required) | Available full-time May 2027

SUMMARY

Mechanical design engineer (MS, May 2027) with **3+ years of SolidWorks/Fusion 360** parts, assemblies, GD&T drawings and BOMs, plus hands-on CNC, sheet metal and assembly. Designed an 8-part AHSS enclosure for a Ford Mach-E (160 L of cells, 68 fasteners) that held a **100 kN side-pole load at a minimum safety factor of 1.5** in FEA, and specified die angles that compensate for spring back. Led a 5-member team to design, 3D print and wind-tunnel test a wing, **cutting 25% material** with a dovetail split joint and reaching **90% agreement with theory**. Delivered a 20+ part SolidWorks prototype with 10 drawings at Cupp Golf; built **15 autonomous vehicles** at MIT; waterjet-cut **30+ race-car parts**.

SKILLS

- **CAD & Drafting:** SolidWorks (parts, assemblies, drawings, sheet metal), Fusion 360, AutoCAD, GD&T (ASME Y14.5), BOMs, DFM/DFA.
- **Fabrication:** CNC waterjet, shop machining, sheet metal, 3D printing, soldering/PCB assembly, electromechanical assembly.
- **Analysis & Software:** Ansys Mechanical & Fluent, Fusion 360 FEA, hand calcs, MATLAB, Python, Git, Microsoft Office.

WORK EXPERIENCE

CAD Intern | Cupp Golf, Los Angeles, CA (Remote) *Sep 2026 – Present*
- Modeled a 20+ component Radius Golf Aid prototype assembly in SolidWorks; used mates to check motion and interference.
- Released 10 detail/assembly drawings with manufacturing notes and revision tables, and a BOM of 15+ McMaster-Carr parts.
- Reviewed design safety, structural integrity and commercial viability with the founders over 10+ weeks.

Autonomous RACECAR Teaching Assistant | MIT Beaver Works Summer Institute, Cambridge, MA *Jun 2026 – Aug 2026*
- Built and maintained 15 RACECAR Neo autonomous vehicles: 3D-printed mounts, bolted/magnetic attachments, cable routing.
- Soldered custom power-distribution PCBs, integrated Teensy/Raspberry Pi controls and sensors, and debugged fleet faults.
- Taught PID control and test procedures to high-school students and U.S. Air Force pilots.

CFD Engineering Intern | Flowthermolab Ltd, Nottingham, UK (Remote) *Jan 2025 – Aug 2025*
- Produced 5+ GD&T drawings of a globe valve assembly and prepared its CAD model for Ansys Fluent meshing.
- Simulated flow at valve openings of 1–20 mm, quantifying a ~20% decrease in pressure drop to support valve sizing.

Waterjet Technician | Salford Racing (Formula Student), Salford, UK *Jan 2023 – Jul 2023*
- Waterjet-cut 30+ metal and carbon-fiber chassis, exhaust and structural parts to drawing and H&S procedures.
- Prepared 20+ part cut files in Illustrator; built 2 shadow boards, improving workshop efficiency by 15%.

ENGINEERING PROJECTS

EV Battery Enclosure Design | Build Fellowship / Open Avenues | *Fusion 360, FEA, GD&T, BOM* *Feb 2026 – Apr 2026*
- Designed an 8-component AHSS sheet-metal enclosure (2062 × 1343 × 128 mm) with cooling plates and 68 M6 fasteners.
- FEA of 100 kN side-pole and 19.6 kN underfloor loads (FMVSS 305 / ECE R100), min. safety factor 1.5; verified by hand calcs.
- Specified roll forming, an 81.28° die angle for spring back (K = 0.984), and multi-pulse spot welds to limit LME.

Transonic Wing Prototyping & Wind Tunnel Testing | *SolidWorks, 3D printing, XFOIL* *Jan 2024 – Apr 2024*
- Led a 5-member team to design a 0.5 m span lofted wing in SolidWorks with 10 pressure taps built into the geometry.
- Split the wing with a dovetail joint to fit the printer, cutting material 25% through prototype iteration.
- Tested at 5–25 m/s, located boundary-layer separation, and reached 90% agreement with theory/XFOIL.

Screw Gauge Manufacturing | Trafford College, Manchester, UK *Oct 2023*
- Machined a mild-steel screw gauge from GD&T blueprints using hand tools and shop machinery.

EDUCATION

Illinois Institute of Technology, Chicago, IL | MS, Mechanical & Aerospace Engineering | GPA 3.8/4.0 *May 2027*
- Commencement Honor Marshal; 40% merit-based tuition scholarship.

University of Salford, Salford, UK | BEng (Hons) Aeronautical Engineering, First Class Honours *Jun 2024*
- Ranked 2nd in the program; awarded the Gold Excellence Scholarship.

LEADERSHIP

Technical Project Director | AIAA at Illinois Tech, Chicago, IL *Sep 2025 – Present*
- Run weekly Fusion 360 CAD workshops for 30+ members and lead 10+ executive board meetings.